

2 Bees are insects that produce venom as a means of self-defence.

Melittin is a polypeptide present in the venom of bees.

(a) Lysine is one of the amino acids present in melittin.

Fig. 2.1 shows an incomplete diagram of the structure of lysine.

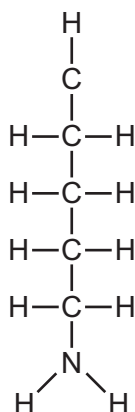


Fig. 2.1

(i) State the number of carbon atoms in the R group of lysine.

..... [1]

(ii) Complete the diagram of the structure of lysine in Fig. 2.1. [2]

(b) Descriptions of the structure of melittin are shown in Table 2.1.

Complete Table 2.1 by writing the level of protein structure that applies to each description.

Table 2.1

description	level of protein structure
in some conditions, four melittin polypeptides can bind to each other	
a melittin polypeptide consists of a sequence of 26 amino acids	
alpha helices are formed at each end of a melittin polypeptide	

[3]

- (c) When a bee stings a person, venom enters the body.

Melittin in the venom interacts with cell surface membranes of body cells, as shown in Fig. 2.2.

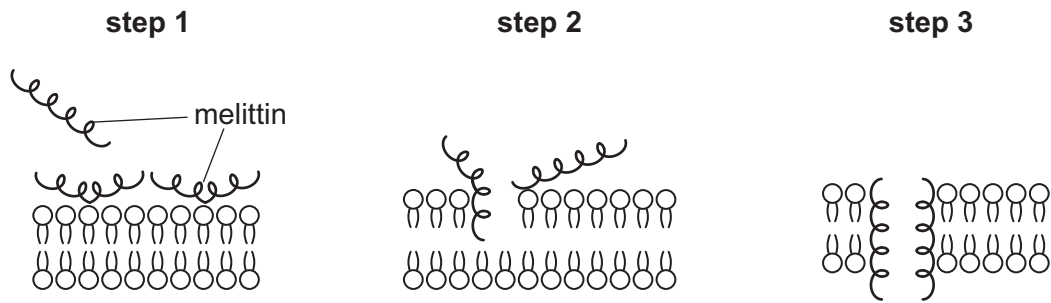


Fig. 2.2

Outline how melittin affects the structure of a cell surface membrane, as shown in Fig. 2.2.

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..... [2]

- (d) Cells that have been affected by melittin break down into cell fragments.

These cell fragments are taken in by phagocytes for further breakdown.

Describe the process by which phagocytes take in and break down these cell fragments.

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..... [4]

[Total: 12]